

Tech & Teach

Digital Chip Design Program

RTL Design

Roaa ALnawshan

Rawan aloyayri

Renad aloyayri

Lana alsuhaibani

Lara alsuhaibani

Rand Almutlaq

Tech & Teach

Task#10

22/7/2026

Seq & Comb Circuits Datapath – Part 1

Version 1.0

Task #1.1

```
module mem #(  
    parameter int addr_width = 5,  
    parameter int data_width = 8  
)  
(  
    input logic clk,  
    input logic write,  
    input logic [addr_width-1:0] addr,  
    input logic [data_width-1:0] data_in,  
    output logic [data_width-1:0] data_out  
);  
  
    localparam int depth = 1 << addr_width;  
  
    logic [data_width-1:0] memory [0:depth-1];  
  
    always_ff @(posedge clk) begin  
        if (write) begin  
            memory[addr] <= data_in;  
        end  
    end  
  
    assign data_out = memory[addr];  
  
endmodule
```

Tech & Teach

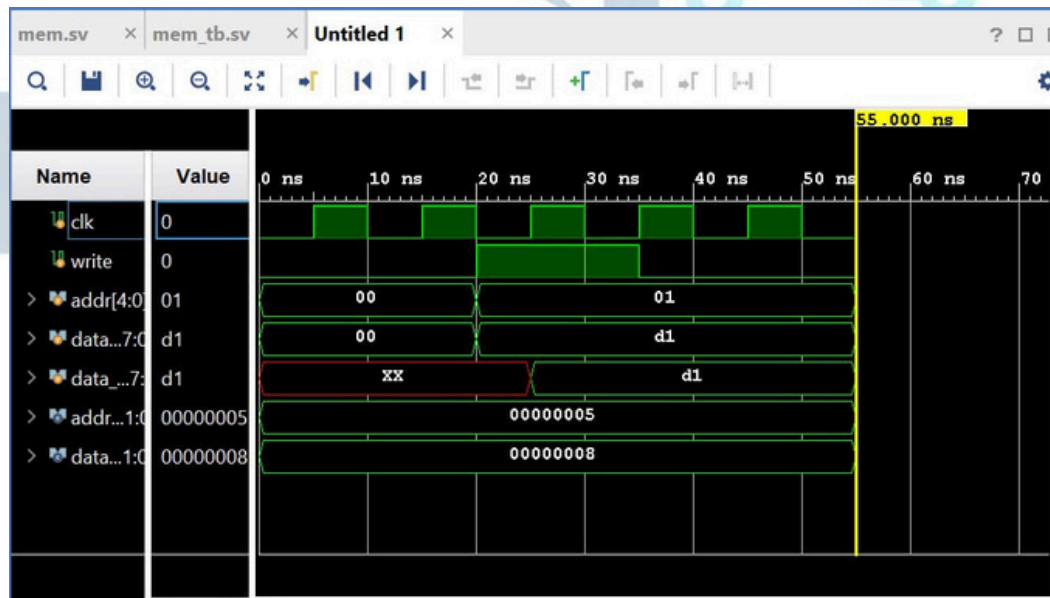
Test bench

```
`timescale 1ns / 1ps

module mem_tb;
parameter addr_width = 5;
parameter data_width = 8;
logic clk;
logic write;
logic [addr_width-1:0] addr;
logic [data_width-1:0] data_in;
logic [data_width-1:0] data_out;

mem #(
    .addr_width(addr_width),
    .data_width(data_width)
)
dut (
    .clk(clk),
    .write(write),
    .addr(addr),
    .data_in(data_in),
    .data_out(data_out)
);

always #5 clk=~clk;
initial begin
    clk=0;
    write=0;
    addr=0;
    data_in=0;
    #20;
    write=1;
    addr=1;
    data_in=8'hD1;
    @(posedge clk);
    #10;
    write=0;
    addr=1;
    #20;
    $finish;
end
endmodule
```



Work distribution

Section	ROAA	RAND	LANA	LARA	RAWAN	RENAD
Report			✓	✓		
Task 1.1	✓	✓			✓	✓